Chapter 1: The Measurement of Interest

Interest may be defined as the compensation that a borrower of capital pays to a lender of capital for its use.

Principal is the initial amount of money invested.

Accumulated amount The total amount of money received after a period of time.

The amount of **interest** earned is the difference between the accumulated amount and the principal.

The **period** is the unit in which time is measured. The more common measurement period is one year, and will be assumed unless otherwise stated.

The **accumulation function,** denoted **a(t)**, gives the accumulated value at time $t \geq 0$ of an original investment of 1. The properties of $a(t)$ are:

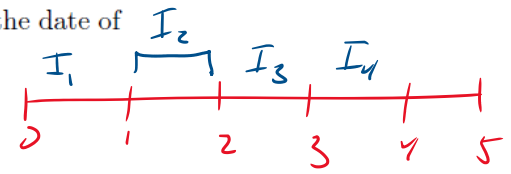
- $a(0) = 1$
- For positive interest rates, $a(t)$ in an increasing function. Negative interest rates gives a decreasing function.
- If interest accrues continuously, the function will be continuous. Otherwise, $a(t)$ will have discontinuities.

The **amount function,** denoted **A(t)**, gives the accumulated value at time $t \geq 0$ of an original investment of k. The properties of $A(t)$ are:

- $A(t) = k * a(t)$ so we get $A(0) = k$
- For positive interest rates, $A(t)$ in an increasing function. Negative interest rates gives a decreasing function.
- If interest accrues continuously, the function will be continuous. Otherwise, $A(t)$ will have discontinuities.

Define I_n as the amount of interest earned in the n-th period from the date of investment.

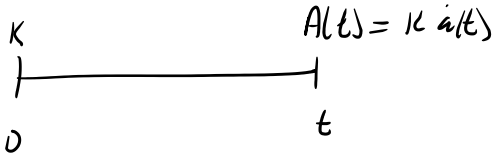
$$I_n = A(n) - A(n-1) \text{ for } n = 1, 2, 3, \cdots$$



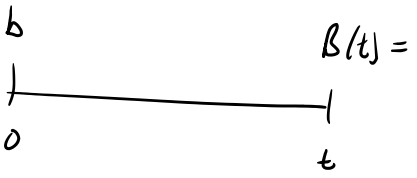
$$I_2 = A(2) - A(1)$$

Proportionality: Suppose an investment of b is made and this investment will follow another investment strategy whose amount function is given by $A(t)$, $t > 0$. Assume no other deposits or withdrawls for this investment.

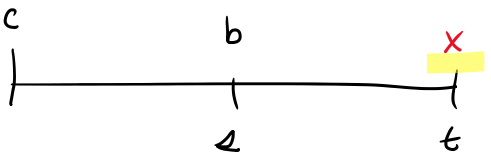
- If the investment of b is made at time 0, then the value of the investment at time t is given by $\dfrac{b \cdot A(t)}{A(0)}$.



$$B(t) = b\,a(t) = b \cdot \frac{K a(t)}{K} = \frac{b\,A(t)}{A(0)}$$



- If the investment of b is made at time s, then the value of the investment at time t is given by $\dfrac{b \cdot A(t)}{A(s)}$.



Let $C = B(0)$ $\qquad B(s) = b = C\,a(s)$ $\qquad C = \dfrac{b}{a(s)}$

$$X = B(t) = C\,a(t) = \frac{b}{a(s)}\,a(t) = b\,\frac{a(t)}{a(s)} = b\,\frac{K a(t)}{K a(s)} = \frac{b\,A(t)}{A(s)}$$

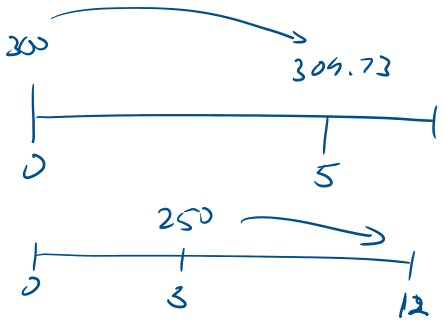
Example: An investment of \$10,000 is made into a fund at time $t = 0$. The fund develops the following balances over the next 4 years.

t	$A(t)$	I_n
0	10,000	
1	10,600	$I_1 = 600$
2	11,024	$I_2 = 424$
3	11,575.2	$I_3 = 551.2$
4	12,732.72	$I_4 = 1{,}157.52$

If \$5,000 is invested at $t = 1$ under the same interest environment, find the accumulated value of the \$5,000 at time $t = 3$.

$$\frac{5000\ A(3)}{A(1)} = \frac{5000\ (11575.2)}{10600} = 5460$$

Example: It is known that $a(t)$ is of the form $a(t) = be^{0.1t} + c$. If \$300 invested at time 0 accumulates to \$309.73 at time 5, find the accumulated value at time 12 of \$250 invested at time 3.



$A(t) = 300\, a(t)$

$A(5) = 309.73$

$309.73 = 300\left(be^{.5} + c\right)$

$\frac{309.73}{300} = be^{.5} + 1 - b$

$\frac{309.73}{300} - 1 = b\left(e^{.5} - 1\right)$

$b = \frac{\frac{309.73}{300} - 1}{e^{.5} - 1} = .049995 \approx .05$

$c = 1 - b = .95$

$a(0) = 1$

$1 = be^{0} + c$

$1 = b + c$

$c = 1 - b$

$\frac{250\, A(12)}{A(3)}$

$a(t) = .05e^{.1t} + .95$

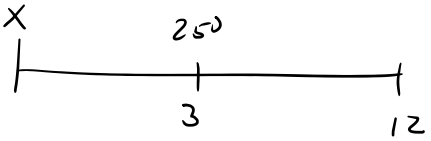
$A(t) = 300\left(.05e^{.1t} + .95\right)$

$\text{Answer} = \frac{250\, A(12)}{A(3)}$

$= \$274.20$

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method 2



find $a(t)$

$B(t) = X\, a(t)$

$B(3) = 250$ to solve for X

now find $B(12)$

Section 1.3: The Effective Rate of Interest

The **effective rate of interest**, i, is the amount of money that one unit invested at the beginning of a period will earn during the period, where interest is paid at the end of the period. This assumes the principal remains constant throughout the period.

$a(0) = 1$

$a(1) = 1 + i$

$i = a(1) - a(0)$

Note: The effective rate of interest is often expressed as a percentage. Thus $7\% \equiv 0.07$ earned per unit of principal.

The effective rate of interest for a period can be defined in terms of the amount function by the following.

$$i = a(1) - a(0) = \frac{a(1) - a(0)}{a(0)} = \frac{Ka(1) - Ka(0)}{Ka(0)} = \frac{A(1) - A(0)}{A(0)}$$

The effective rate of interest can be computed during the n-th period.

$$i_n = \frac{A(n) - A(n-1)}{A(n-1)} = \frac{I_n}{A(n-1)}$$

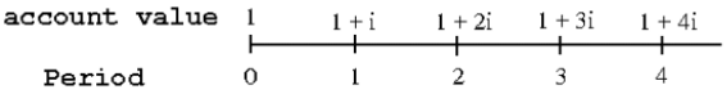
The eff. Rate of Interest during the nth period is the Ratio of the Interest earned in the nth period to the amt invested at the begining of the nth period.

Example: An investment of \$10,000 is made into a fund at time $t = 0$. The fund develops the following balances over the next 4 years. Find the effective rate of interest for each of the 3 years.

t	$A(t)$	I_n	i_n
0	10,000		
1	10,600	$I_1 = 600$	$i_1 = \frac{600}{10000} = .06 \rightarrow 6\%$
2	11,024	$I_2 = 424$	$i_2 = \frac{424}{10600} = .04 \rightarrow 4\%$
3	11,575.2	$I_3 = 551.2$	$i_3 = \frac{551.2}{11024} = .05 \rightarrow 5\%$

Section 1.4: simple interest

Simple interest is when the amount of interest earned during each period is constant.



Thus $a(t) = 1 + it$ for $t = 1, 2, 3, \cdots$

Note: A constant rate of simple interest does not imply a constant effective rate of interest.

$$i_n = \frac{a(n) - a(n-1)}{a(n-1)} = \frac{1+ni - [1+(n-1)i]}{1+(n-1)i} = \frac{1+ni-(1+(n-1)i)}{1+(n-1)i} = \frac{1+ni-1-ni+i}{1+(n-1)i}$$

$$i_n = \frac{i}{1+(n-1)i}$$

as $n \to \infty$ $\quad i_n \to 0$

Note: Unless stated otherwise, under simple interest it will be assumed that interest is accrued proportionally over fraction periods.

Example: A bank pays a simple interest rate of 2.5% per annum. $2,000 is deposited on January 1, 2004.
(a) Compute the accumulated value on April 1, 2006.
(b) How long until the accumulated amount is $2,230?

1/1/04 → 4/1/06

2 yrs. 3 months. $= 2 + \frac{3}{12} =$

$= 2.25$ yrs

A) $2000\left(1+(.025)(2.25)\right) = 2112.50$

B) $2000\left(1+(.025)t\right) = 2230$

$1 + .025t = \frac{2230}{2000}$

$.025t = \frac{2230}{2000} - 1$

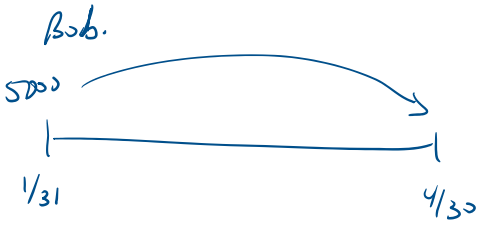
$t = \frac{1}{.025}\left(\frac{2230}{2000} - 1\right) = 4.6$ yrs.

use for 2019

use simple Interest

Example: On January 31 Bob borrows \$5,000 from David and gives David a promissory note. The note states that the loan will be repaid on April 30 of the same year, with interest of 12% per annum. On March 1, David sells the promissory note to Sam, who pays David a sum of money in return for the right to collect the payment from Bob on April 30. Sam pays David an amount such that Sam's yield (interest rate earned) from March 1 to the maturity date can be stated as an annual rate of interest of 15%.

(a) Determine the amount that Bob was to have paid David on April 30.
(b) Determine the amount that Sam paid to David and the simple interest rate David earned on an annual basis.



dbd (1.31.19, 4.30.19) = 89

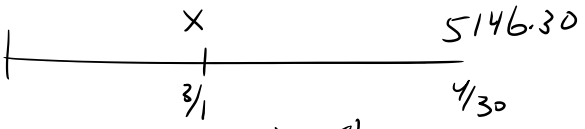
Simple

i = 12% per yr. $t = \frac{89}{365}$

Bob owes $5000\left(1+(.12)\left(\frac{89}{365}\right)\right) = \$5146.30$

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b) Sam



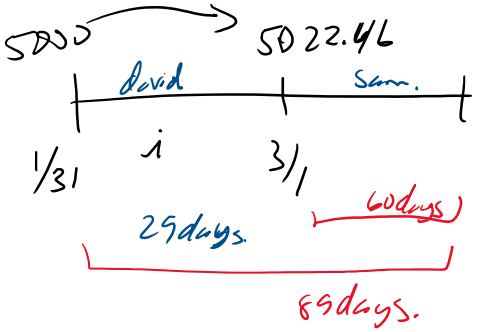
i = 15% per yr simple interest rate

$5146.30 = X\left(1+.15\left(\frac{60}{365}\right)\right)$

$t = \frac{60}{365}$

dbd (3/1/19, 4/30/19) = 60

$X = 5022.46$



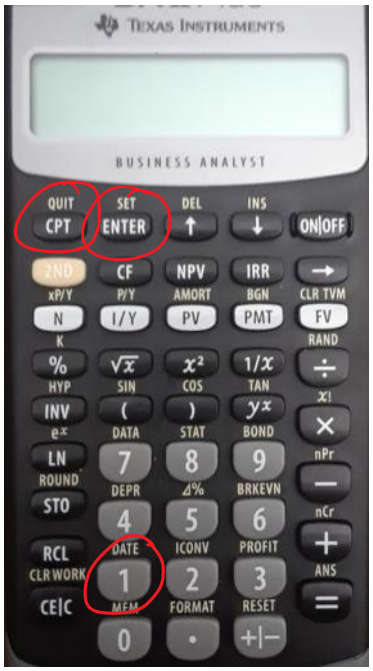
$5022.46 = 5000\left(1+i\frac{29}{365}\right)$

$\frac{5022.46}{5000} = 1+i\frac{29}{365}$

$$i = \frac{365}{29}\left(\frac{5022.46}{5000} - 1\right) = .056537 \rightarrow 5.65\%$$

annual simple Interest Rate.

dbd













① 2nd 1

format mm.ddyy

② enter first date DT1 1.3119

press down arrow

③ enter 2nd date DT2 4.3019

press down arrow twice

④ check for style of calculation: ACT or 360

press up arrow

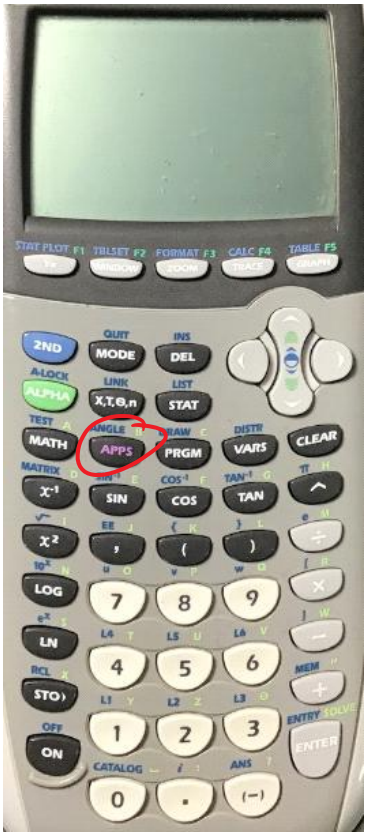
④ for DBD press CPT → 89 days.

↑ actual # of days.

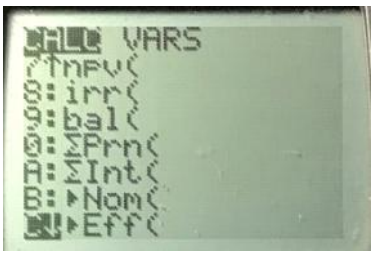
↑ # of days based on a 360 day yr.

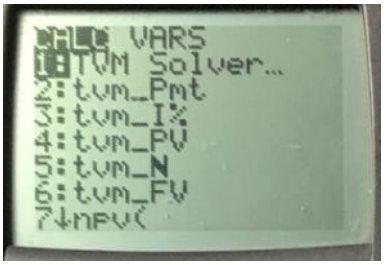
press 2nd enter. to switch for the screens.

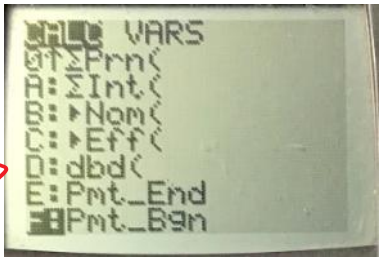
Finance Menu: dbd











mm.ddyy

dbd(1.3119, 4.3019) enter

This method is Always ACT (actual)

Section 1.5: Compound interest

Compound interest is when the interest is automatically reinvested to earn additional interest.

i eff. Rate per period.

t	$a(t)$
0	1
1	$1+i$
2	$(1+i)+i(1+i)$ $=(1+i)(1+i)=(1+i)^2$
3	$(1+i)^2+i(1+i)^2=(1+i)^2[1+i]=(1+i)^3$
⋮	
n	$(1+i)^n$

The accumulation function for $t = 1, 2, 3, \cdots$ is $a(t) = (1+i)^t$

$A(t) = K(1+i)^t$

where i is the compound interest rate for the period.

Note: A constant rate of compound interest implies a constant effective rate of interest.

$$i_n = \frac{a(n) - a(n-1)}{a(n-1)} = \frac{(1+i)^n - (1+i)^{n-1}}{(1+i)^{n-1}} = \frac{(1+i)^{n-1}\left[1+i-1\right]}{(1+i)^{n-1}} = i$$

Note: Unless otherwise stated, assume that interest is accrued over fractional period for compound interest according to the formula.

Example: Find the accumulated value of \$2,000 at the end of 2 years and 3 months invested at 6% per annum.

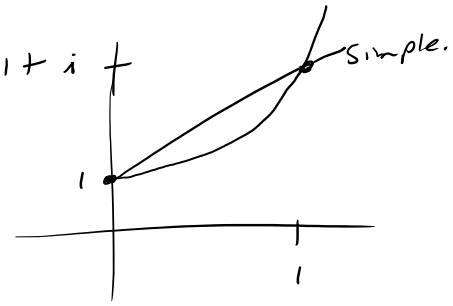
Compound Interest.

$t = 2\text{yr } 3\text{mo} = 2 + \frac{3}{12} = 2.25$

$$2000\left(1+.06\right)^{2.25} = 2280.18$$

Example: Find the accumulated value of \$2,000 at the end of 2 years and 3 months invested at 6% per annum. Assume a simple interest during the final fractional period. Compare this the answer in the previous example.

$$2000\,(1.06)^2\left\{1+(.06)\tfrac{3}{12}\right\} = 2280.91$$



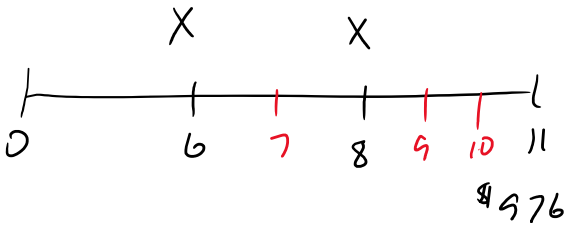
Simple interest is used for mostly for transactions less than one year.

Compound interest is used almost exclusively for financial transactions covering a period of one year or more.

From this point assume compound interest in transactions over 1 year unless told otherwise.

Example: A deposit of \$X is invested at time 6 years at an annual effective interest rate of 8%. A second deposit of \$X is invested at time 8 years at the same interest rate. At time 11 years, the accumulated amount of the investment is \$976. Calculate X.

$i_e = 8\%$ annual.



$$X(1+i)^5 + X(1+i)^3 = 976$$

equation of value.

$$X\left\{(1+i)^5 + (1+i)^3\right\} = 976$$

$$X = \frac{976}{(1+i)^5 + (1+i)^3} = \$357.63$$